Accessible PDFs

In short

PDFs can be made more accessible by turning on tagging, using good contrast, and testing with a screen reader. But they still don't fit different screen sizes, so they're hard to use on phones. Where you can, use a web page or another format instead.

This highlight is about how to make PDFs accessible, and why an alternative is preferred, even after making a PDF accessible.

What is it for?

We use PDF (Portable Document Format) to present all kinds of documents like guidelines, contractual agreements, data-heavy reports, and even site maps. Ideally, all the content in a PDF should be accessible to all, and there are ways to make it accessible. Here are a few ways:

- PDFs are not tagged by default, so make sure to enable tagging so that they can be read by screen readers.
- Maintain good colour contrast
- Use big enough fonts and a one-column layout (this helps people using mobile devices a little, but only a little)
- Add bookmarks to help people move around the document, and set the document title and language in the document properties
- Proper use of headers and footers to provide critical context (document title, section title, and page numbers)
- Using semantically correct tags for the content (just like we do for an accessible HTML page)
- Most important: test it with a screen reader like VoiceOver, and with Adobe Acrobat Pro or a PDF Accessibility Checker (PAC). Make sure every part of the content is accessible.

How to support it

Avoid PDFs, even the accessible ones! Yes, PDFs are not at all as accessible as other formats. For instance, PDFs don't adapt to different screen sizes, so they are hard to use on mobile. They have many other downsides compared with other formats too. So always challenge the requirement of using a PDF and prefer other formats instead of PDFs wherever it's possible. Here are a few alternatives:

- HTML web pages and web applications are the most feature-rich alternatives and can be fully accessible when built with semantic HTML.
- Word documents with accessibility features turned on, and Google Docs or Microsoft Office Online.
    - Offers a built-in accessibility checker.
    - More accessible than PDFs when used properly (heading styles, alt text, and so on)
    - Screen reader-friendly with good keyboard support.
- Accessible Slide Presentations (Google Slides, PowerPoint Online)
    - Can be shared as HTML or accessible PowerPoint format instead of PDF exports.
    - Offers a built-in accessibility checker.
    - Allows embedding of live text, images with alt text, and readable slide order.
- ePub (Electronic Publication Format)
    - Text can reflow to fit different screen sizes.
    - Accessible on e-readers and screen readers.
- Markdown
    - Supports semantic structure and has the ability to provide meaningful alt text for images.
    - Works well with a keyboard and a screen reader

Useful links

- Reading PDFs with reflow and accessibility features
- Adobe: Creating accessible PDFs

Abbreviations

The full form of each abbreviation used in this document.

HTML: HyperText Markup Language

PAC: PDF Accessibility Checker

PDF: Portable Document Format